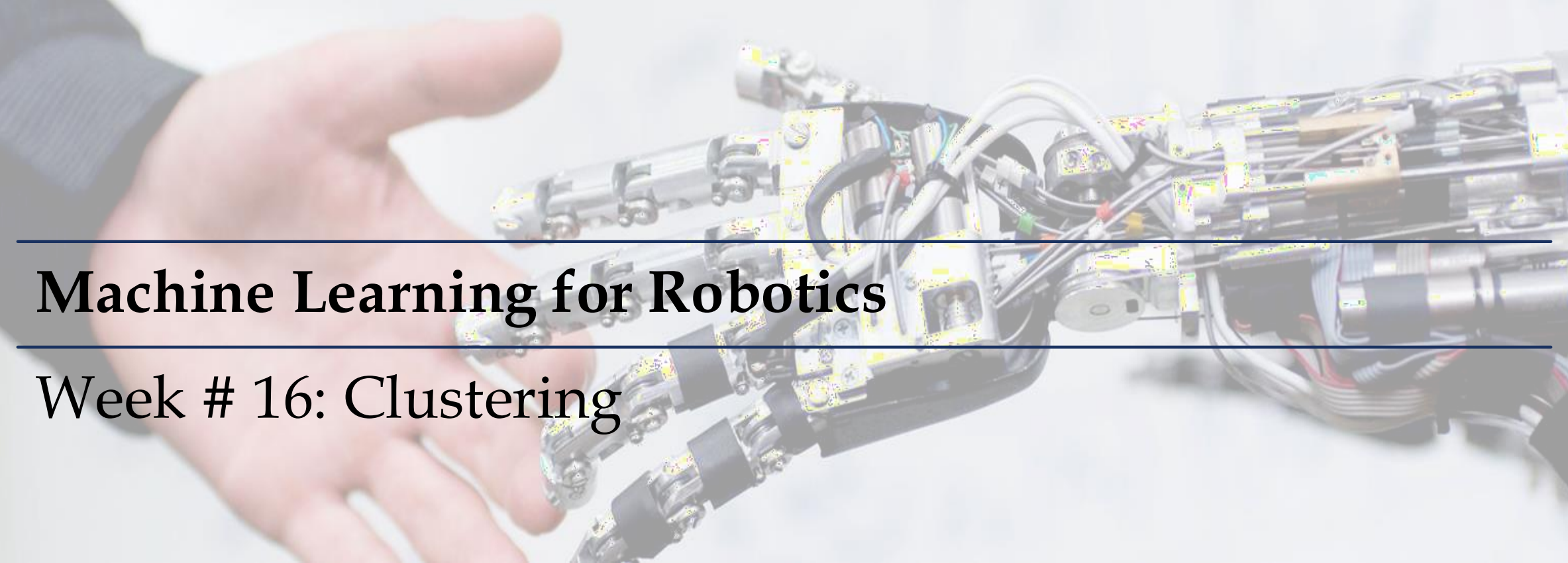


A photograph of a human hand holding a small, intricate robotic arm. The arm is transparent, revealing its internal mechanical and electrical components, including wires, sensors, and small motors. The background is a soft, out-of-focus white.

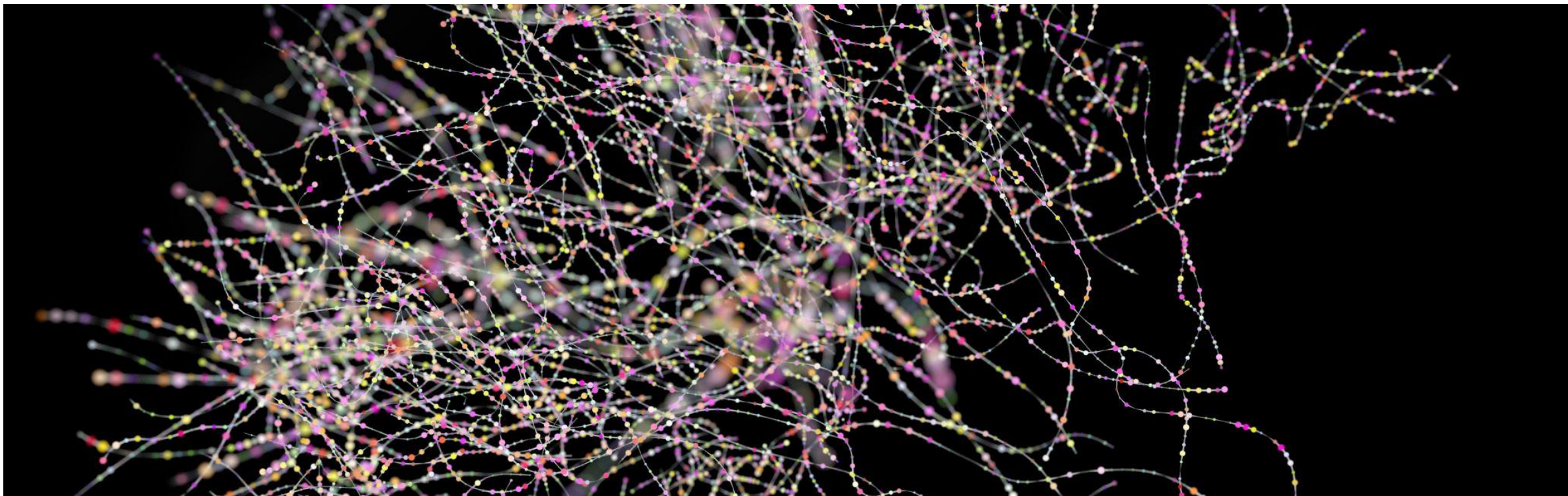
Machine Learning for Robotics

Week # 16: Clustering

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ABOUT INSTRUCTOR

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- Research Publications, Funded Projects and Patents
- Member of AI Task Force (Ministry of Planning, Development & Special Initiatives of Pakistan)
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CLUSTERING

UNSUPERVISED LEARNING



It is basically a type of unsupervised learning method.



An unsupervised learning method is a method in which we draw references from datasets consisting of input data without labeled responses.



Generally, it is used as a process to find meaningful structure, explanatory underlying processes, generative features, and groupings inherent in a set of examples.

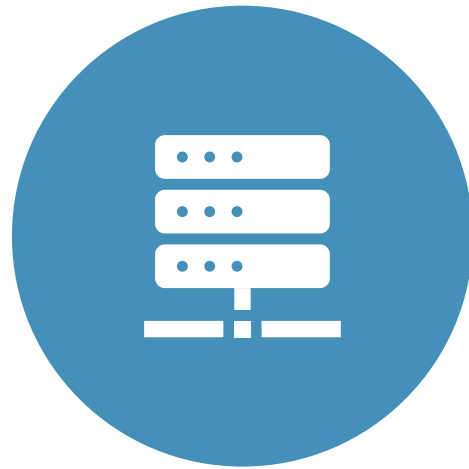
DEFINITION: CLUSTERING

Clustering is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group and dissimilar to the data points in other groups.

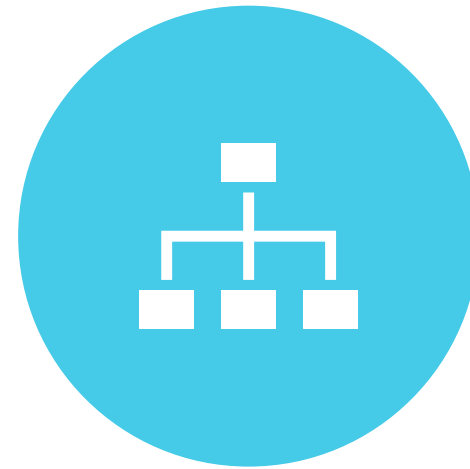


Clustering is very much important as it determines the intrinsic grouping among the unlabelled data present.

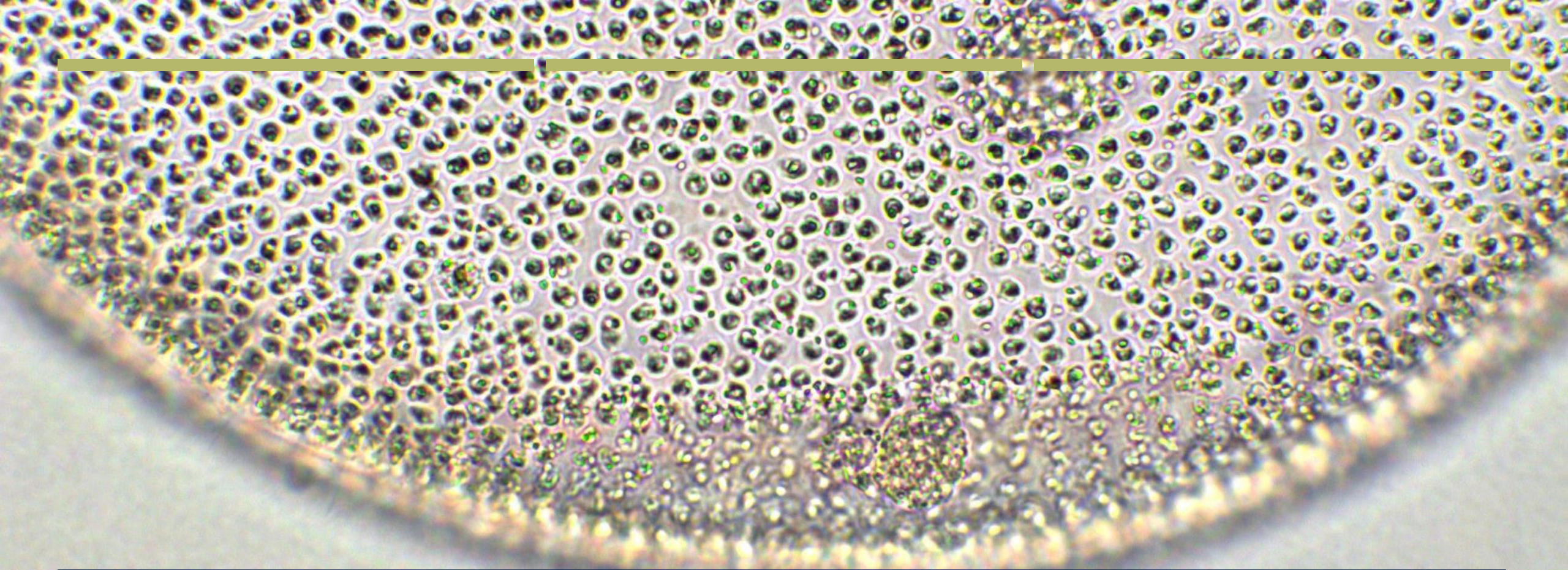
TYPES OF CLUSTERING



K-MEANS CLUSTERING

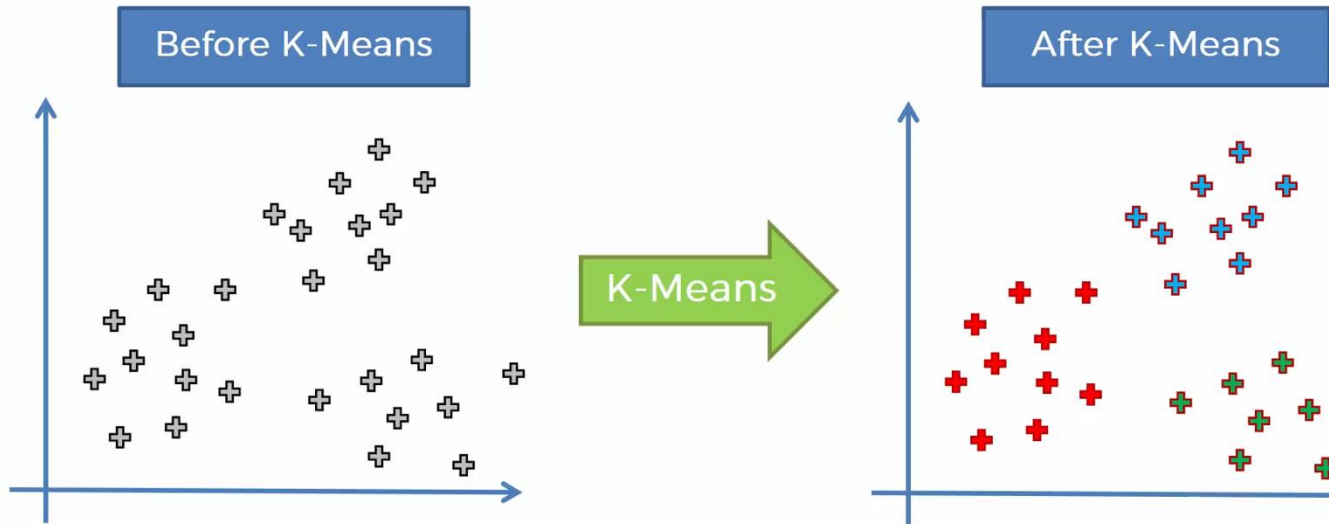


HIERARCHICAL
CLUSTERING



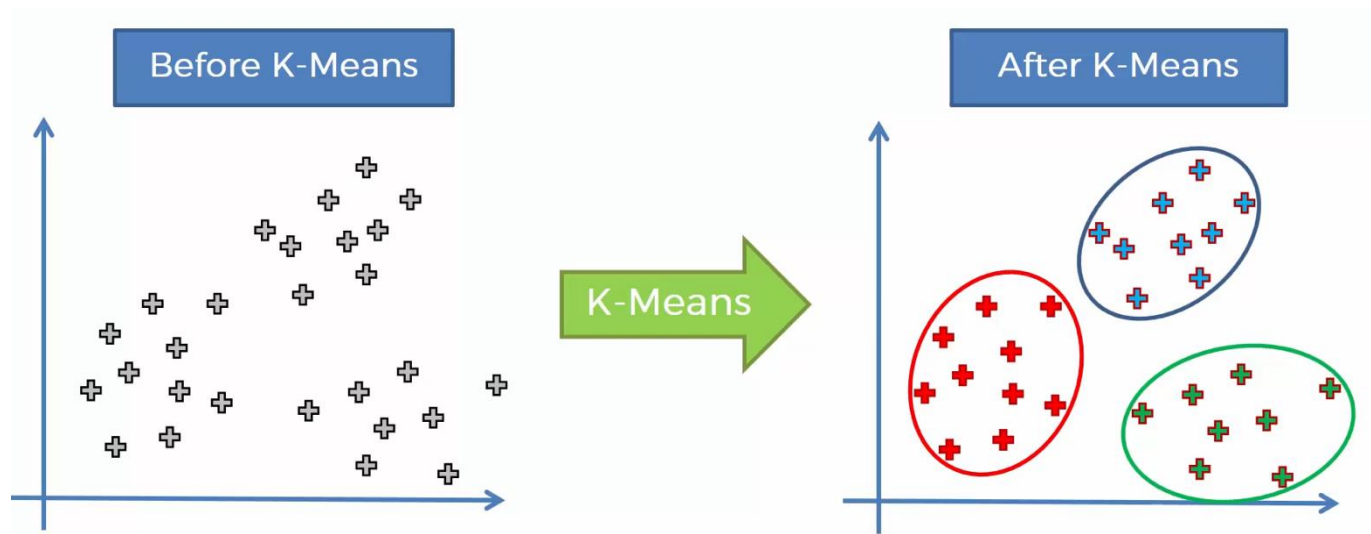
K-MEANS CLUSTERING

What K-Means does for you



- Let us assume that we have a dataset
- The scatter plot is shown in the figure
- We want to find the clusters in the data
- In first look, we can see that there are three clusters in the data.

- We can see that there are three clusters in the data



STEPS FOR K-MEANS CLUSTERING

STEP 1: Choose the number K of clusters



STEP 2: Select at random K points, the centroids (not necessarily from your dataset)



STEP 3: Assign each data point to the closest centroid → That forms K clusters



STEP 4: Compute and place the new centroid of each cluster



STEP 5: Reassign each data point to the new closest centroid.
If any reassignment took place, go to STEP 4, otherwise go to FIN.



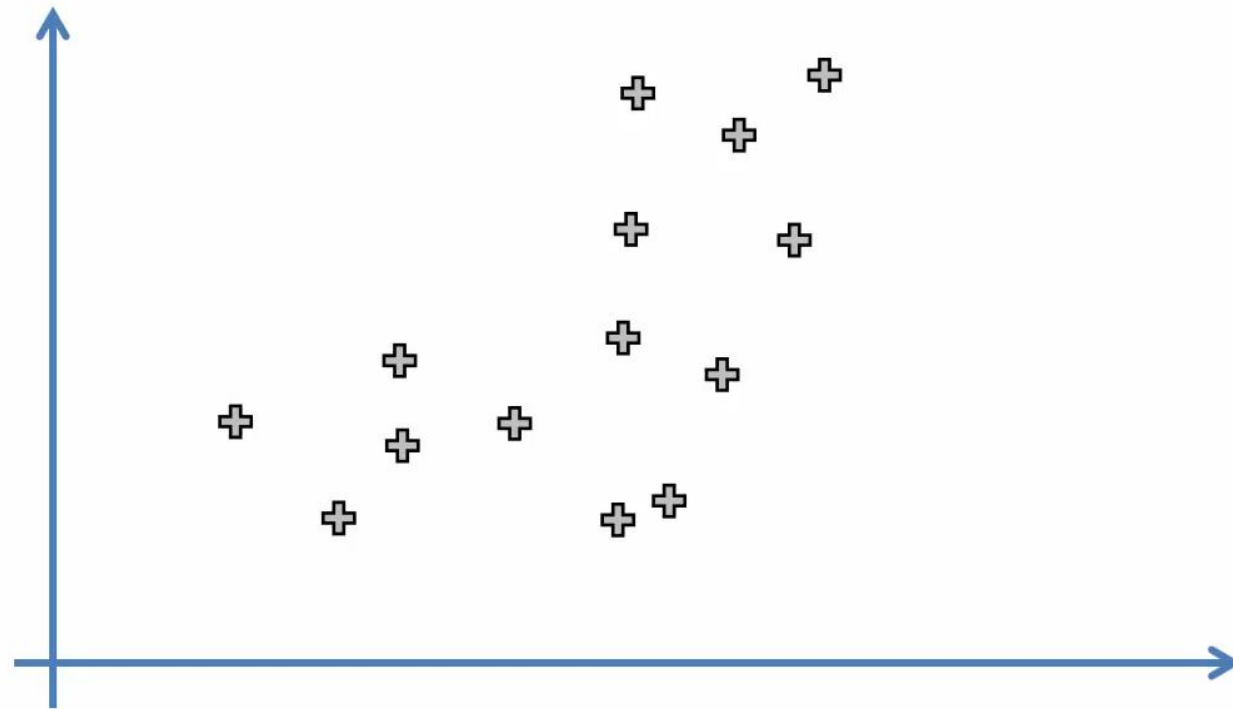
Your Model is Ready

STEP 1: Choose the number K of clusters: $K = 2$

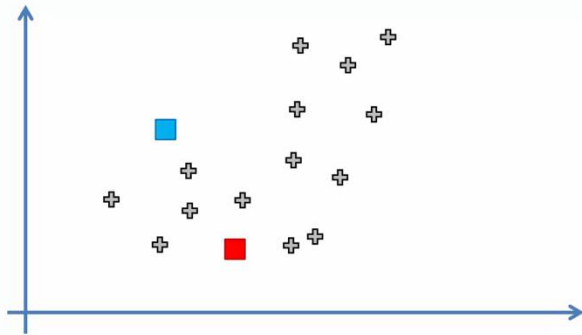


- Can you visually identify the numbers of clusters in the dataset (not very easy!!!)
- Let us assume that we have identified the optimal number of clusters to be 2.

STEP 2: Select at random K points, the centroids (not necessarily from your dataset)

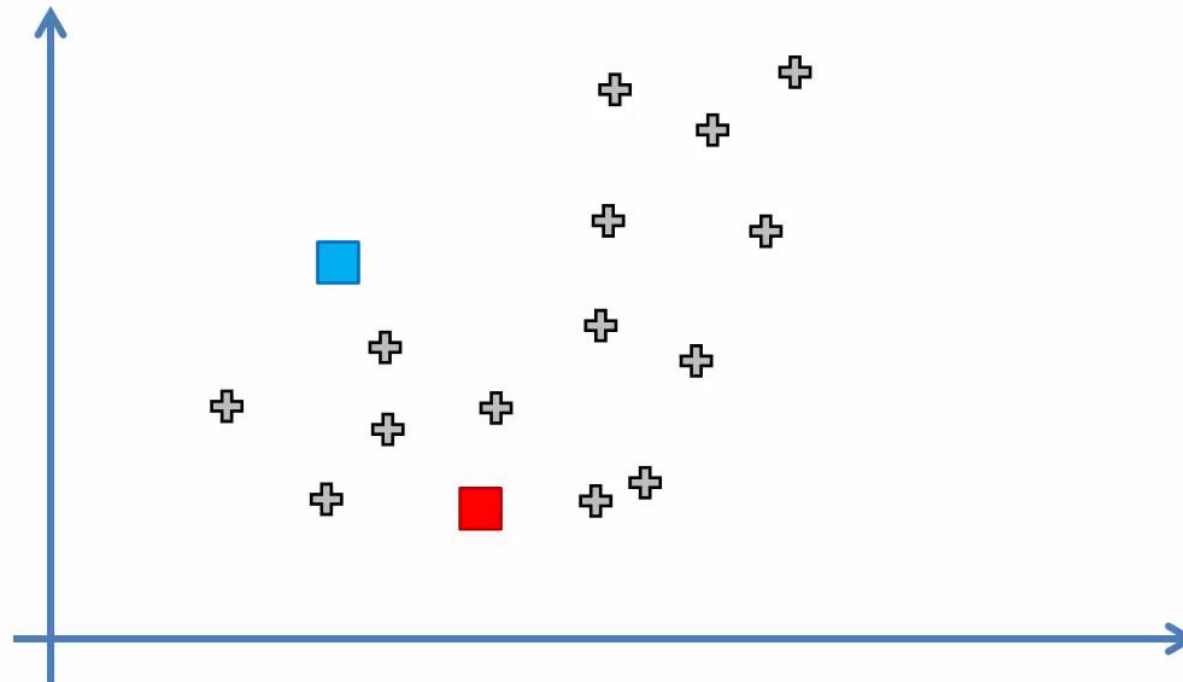


STEP 2: Select at random K points, the centroids (not necessarily from your dataset)

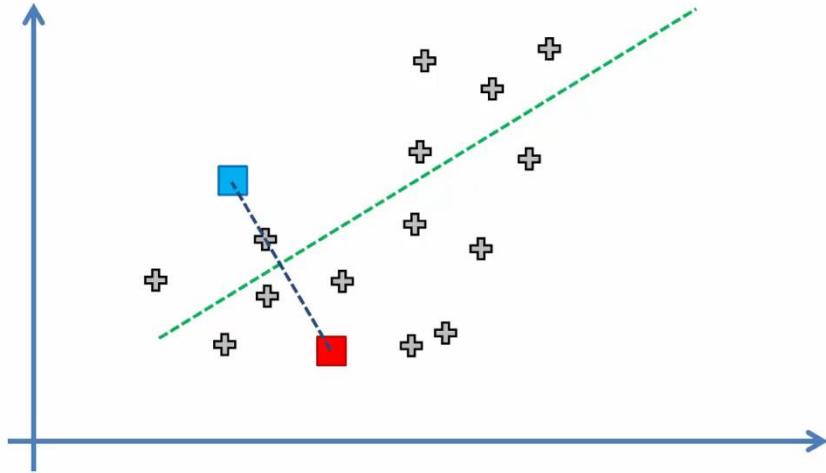


- Let us assume that we select the red and blue points as the centroids

STEP 3: Assign each data point to the closest centroid → That forms K clusters



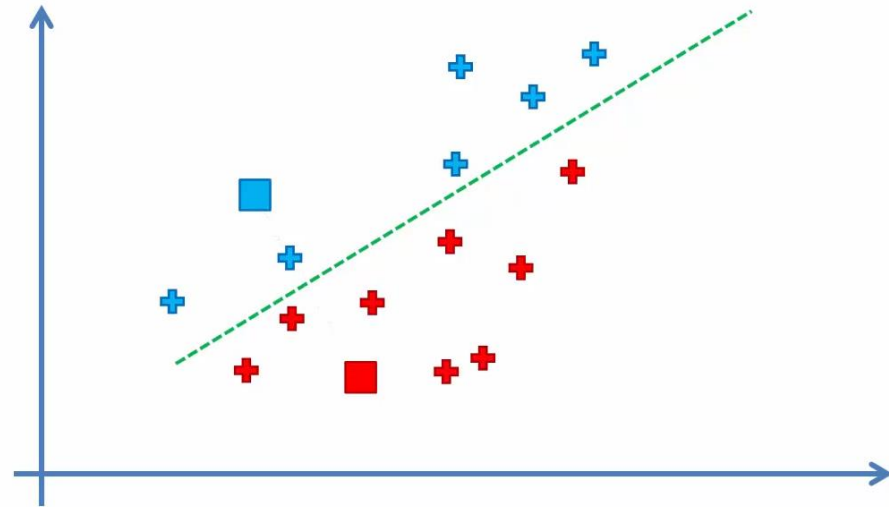
STEP 3: Assign each data point to the closest centroid → That forms K clusters



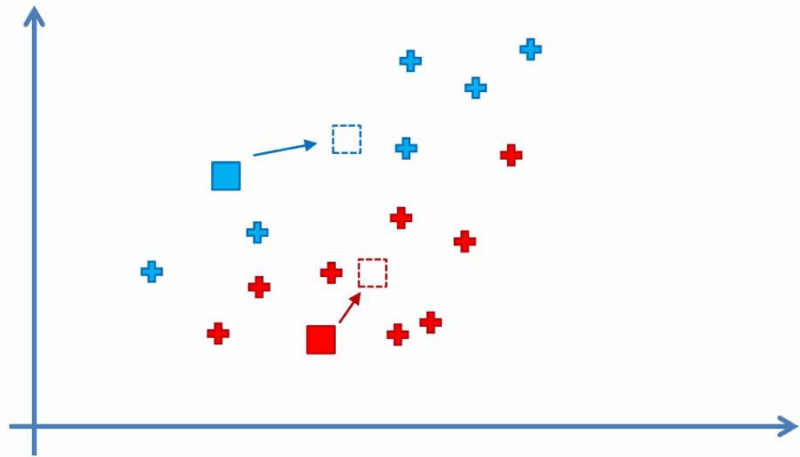
- We know from geometry that the points on the green line are equidistant from the red and blue centroid.
- Now it becomes clear, which points will belong to cluster 1 and cluster 2.

STEP 3: Assign each data point to the closest centroid → That forms K clusters

- Closest centroid is a relative term.
- We are using Euclidean distance here
- But in other scenarios some other parameters may be more appropriate

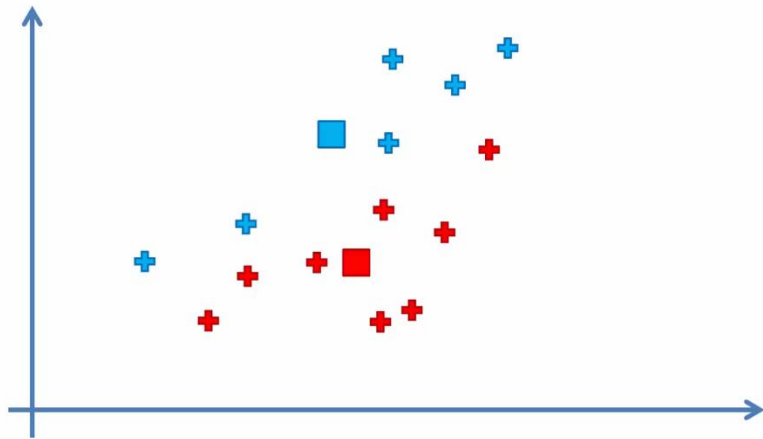


STEP 4: Compute and place the new centroid of each cluster



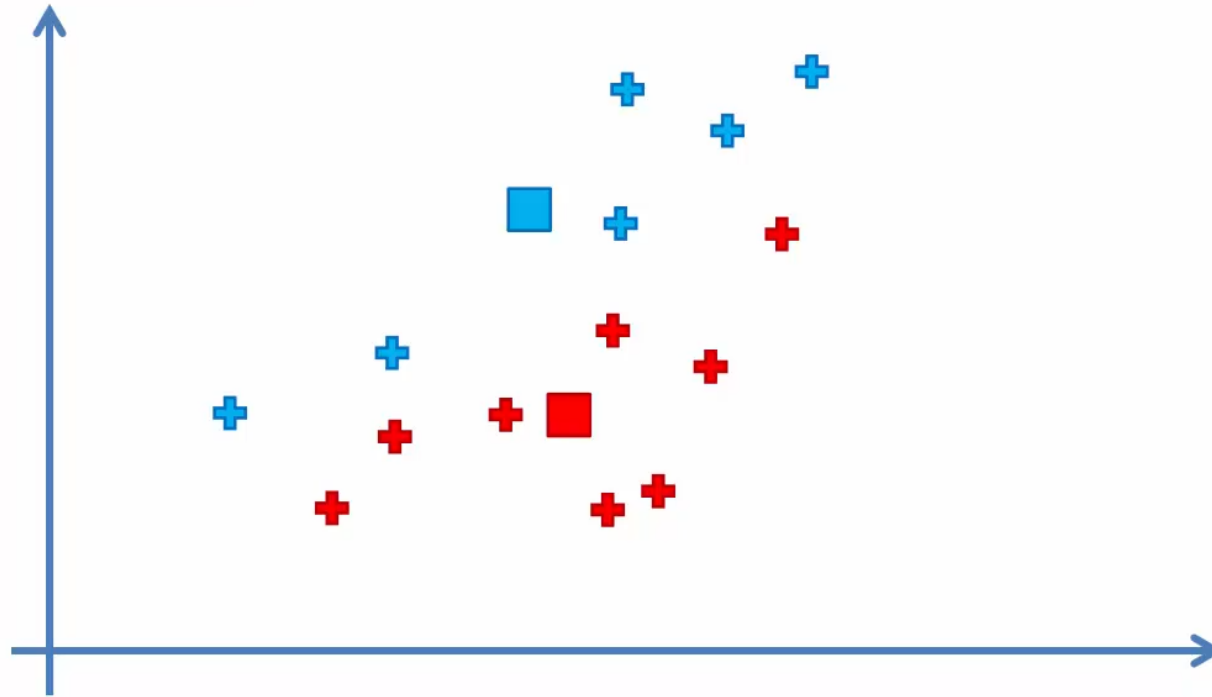
- Compute the new centroids for each cluster, by computing the average/center of gravity of all the points except the centroid itself in each cluster.

STEP 4: Compute and place the new centroid of each cluster

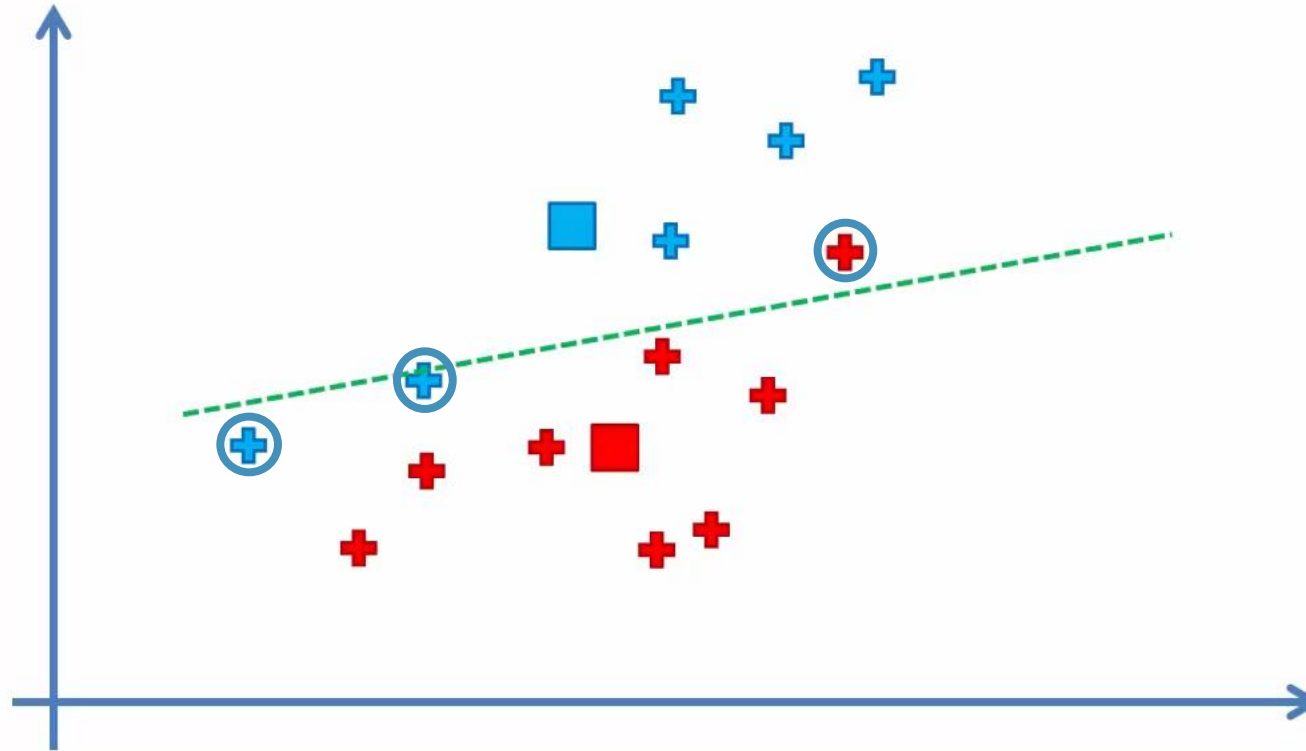


- New centroids have been assigned

STEP 5: Reassign each data point to the new closest centroid.
If any reassignment took place, go to STEP 4, otherwise go to FIN.

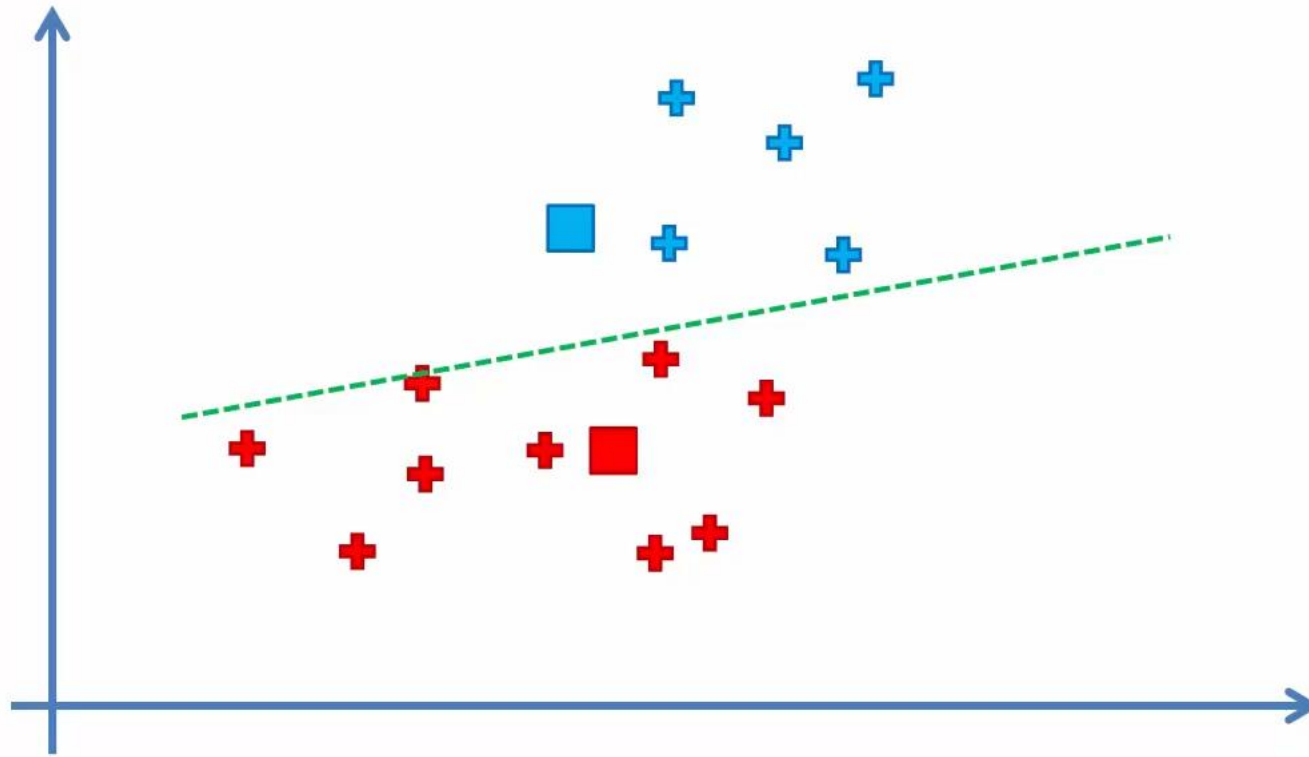


STEP 5: Reassign each data point to the new closest centroid.
If any reassignment took place, go to STEP 4, otherwise go to FIN.



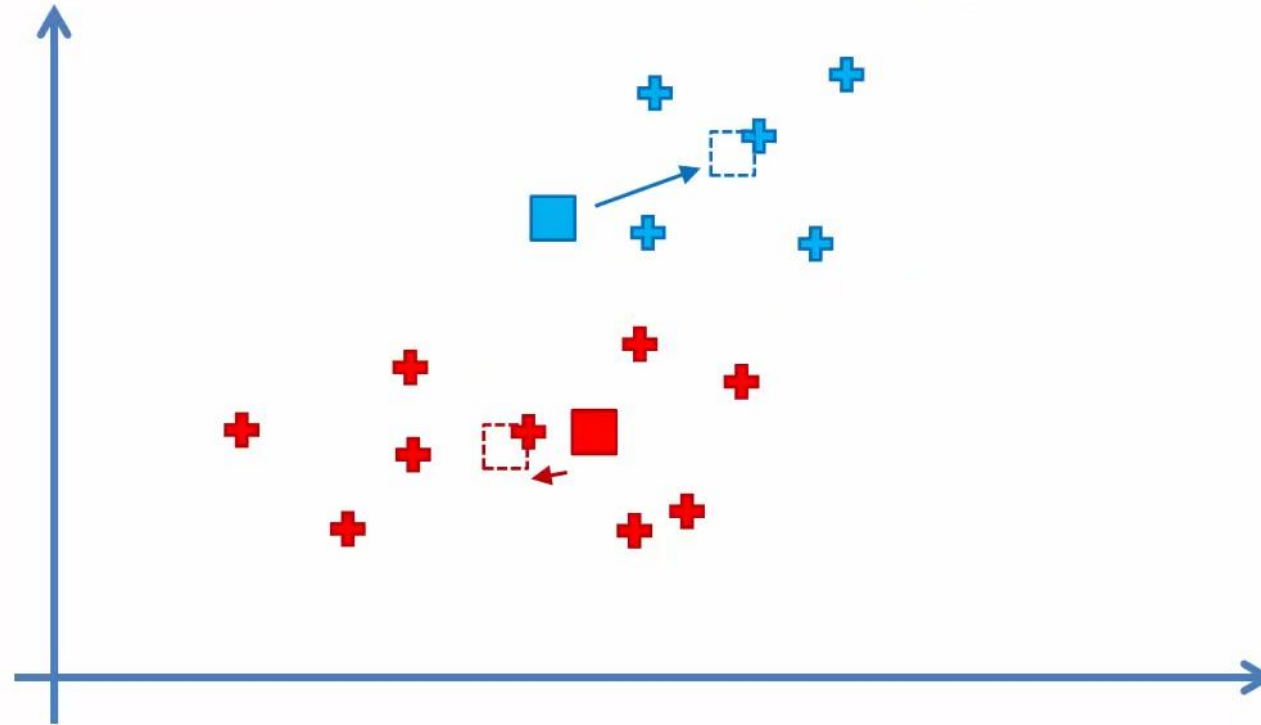
- So, if we plot a line through the scatter plot.
- We can see that three data points are in the wrong cluster

STEP 5: Reassign each data point to the new closest centroid.
If any reassignment took place, go to STEP 4, otherwise go to FIN.



- Now, we will recolor those three points to assign them the correct cluster.
- Since some reassignment as taken place therefore, we go back to step 4

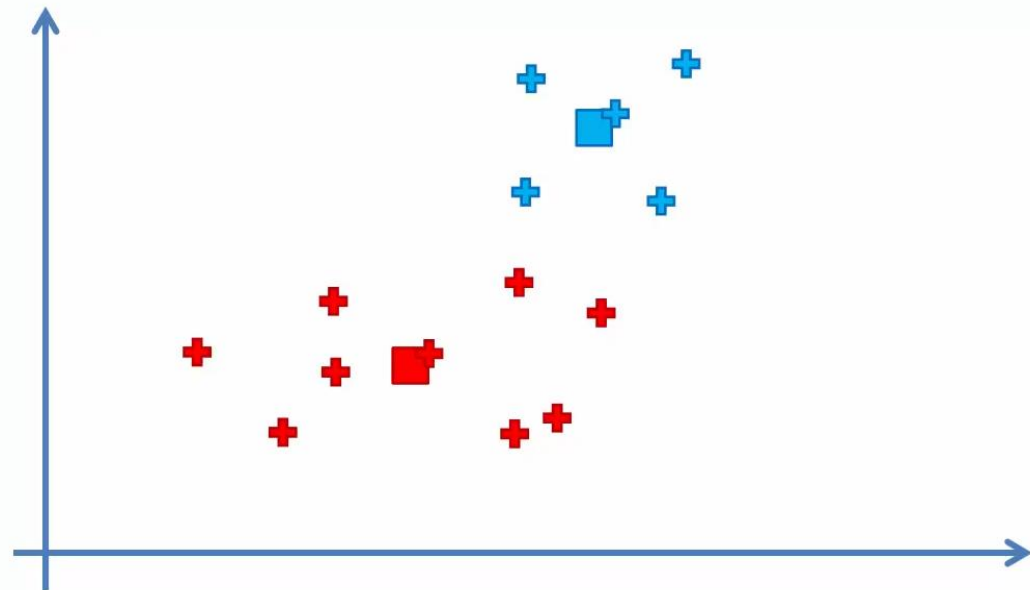
STEP 4: Compute and place the new centroid of each cluster



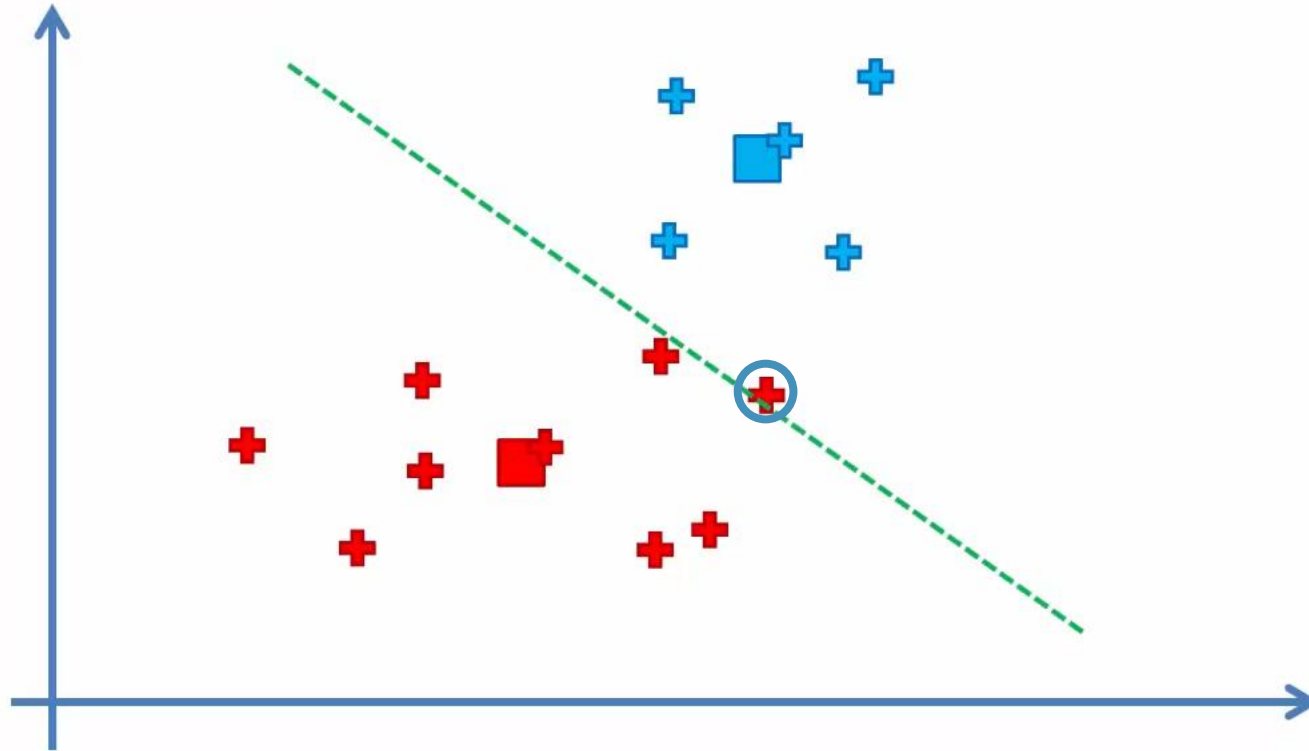
- Compute the center of gravity for the new clusters

- The new centroids have been assigned

STEP 4: Compute and place the new centroid of each cluster

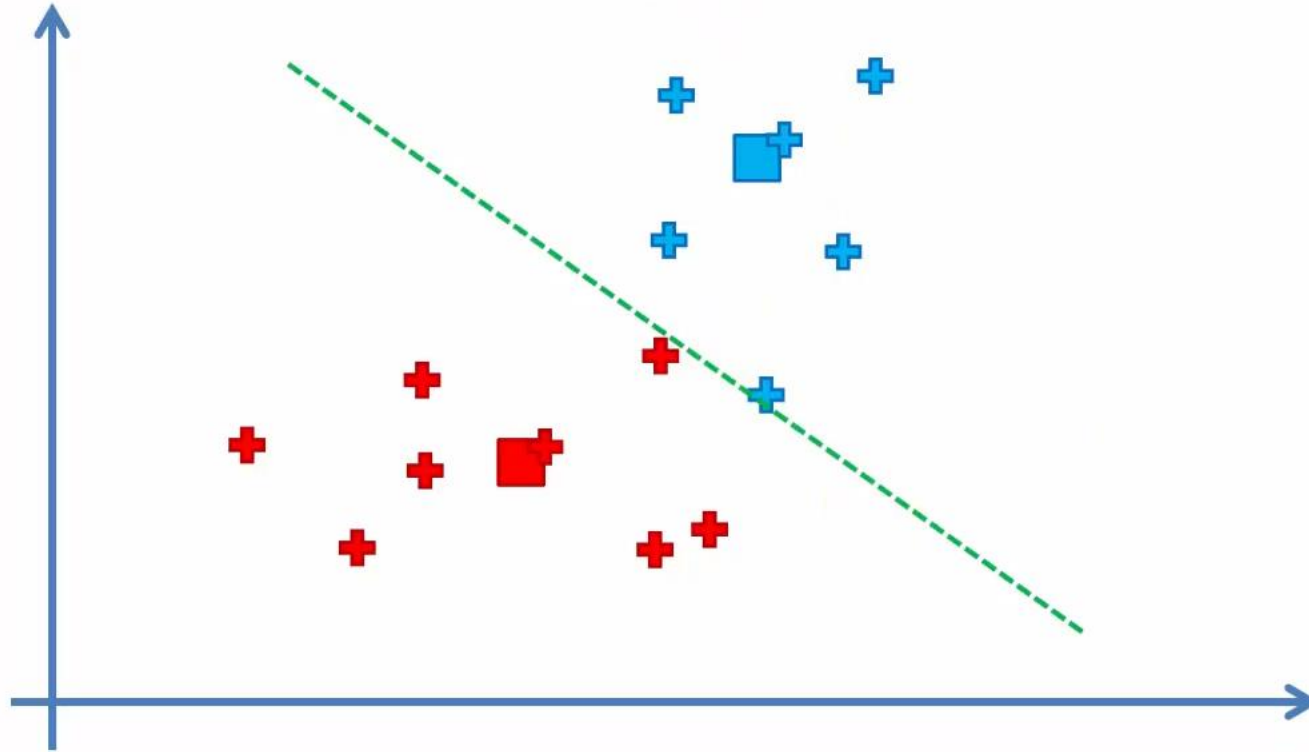


STEP 5: Reassign each data point to the new closest centroid.
If any reassignment took place, go to STEP 4, otherwise go to FIN.



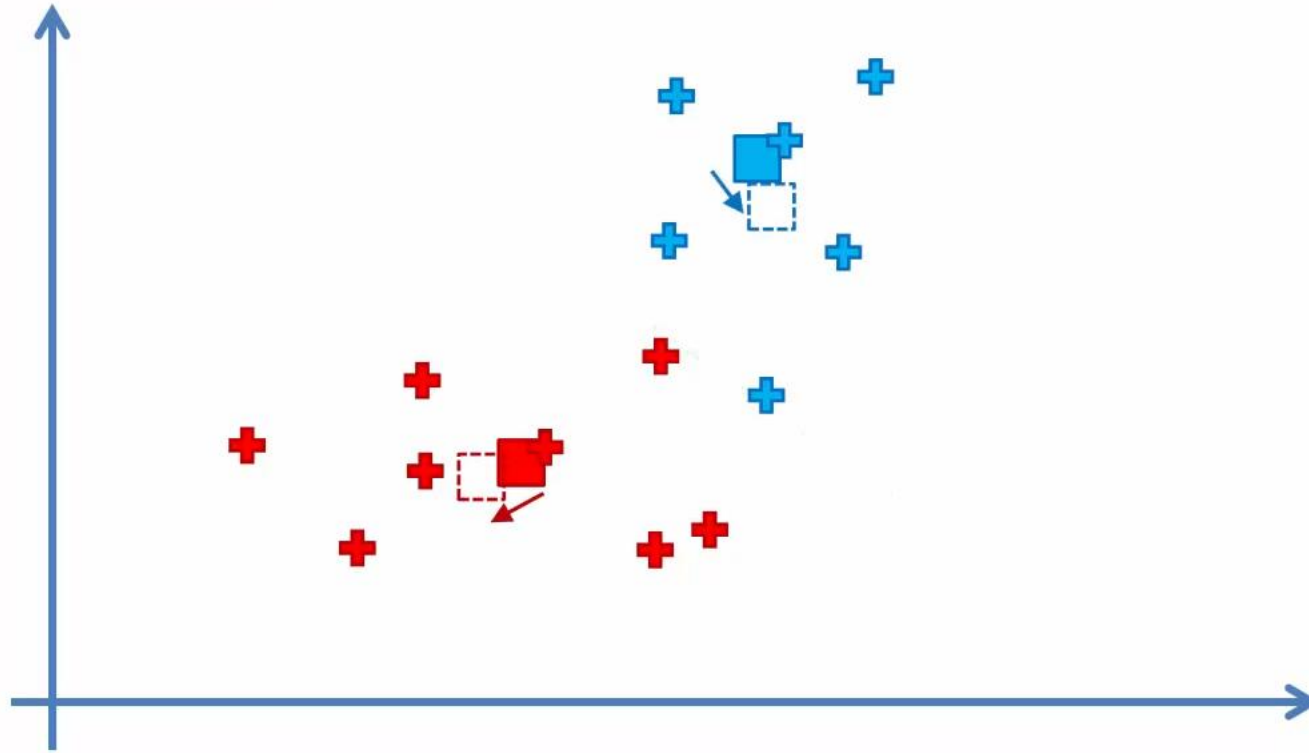
- We again draw the line to check whether any data points are in the wrong cluster
- We see that there is only one point that is in the wrong cluster

STEP 5: Reassign each data point to the new closest centroid.
If any reassignment took place, go to STEP 4, otherwise go to FIN.

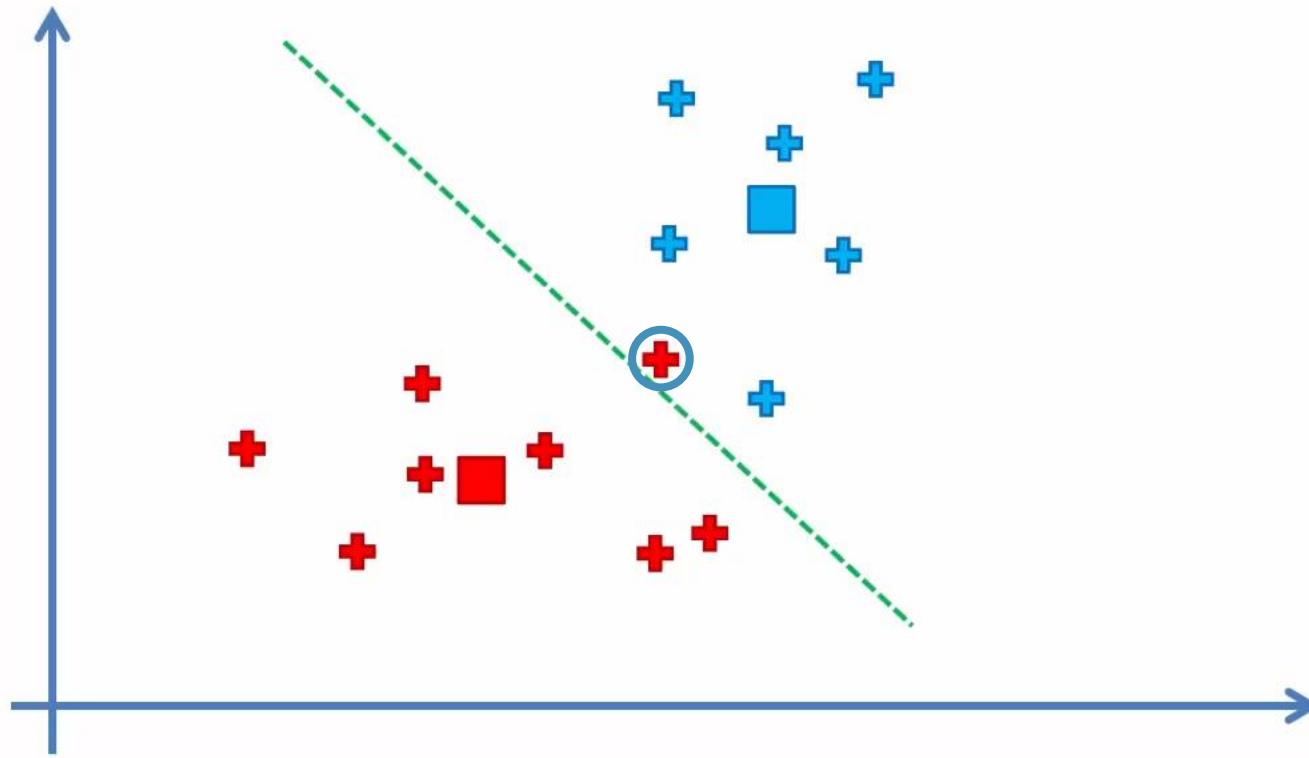


- The point has been reassigned to the blue cluster

STEP 4: Compute and place the new centroid of each cluster

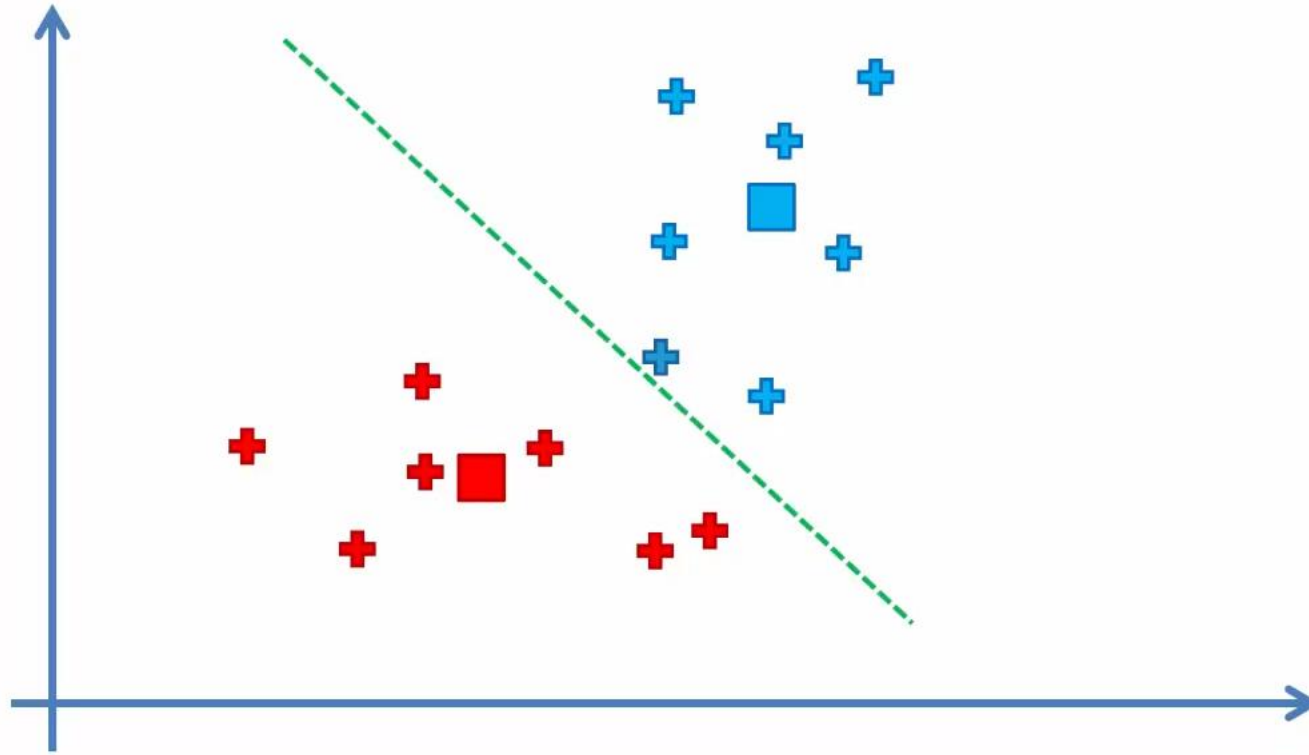


- Next we need to recompute the centroids



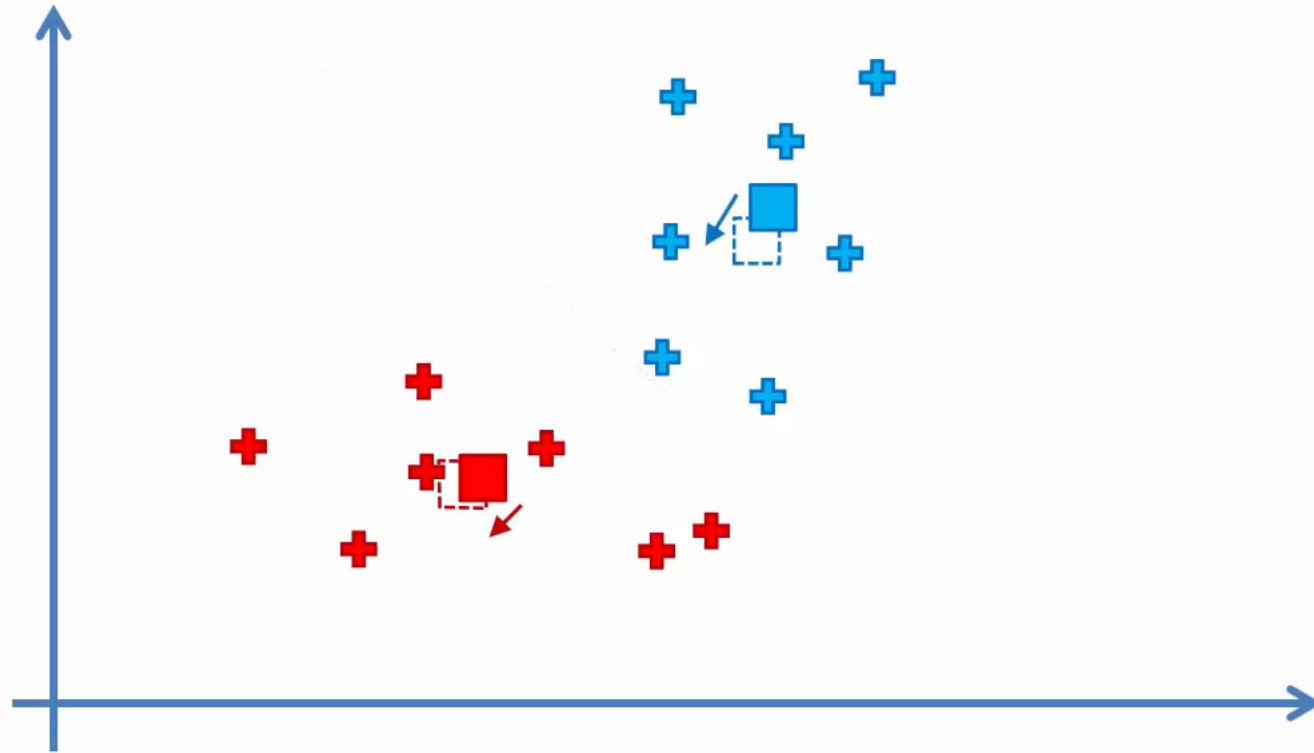
- The centroids have been relocated
- Now only one point needs to be reassigned

STEP 5: Reassign each data point to the new closest centroid.
If any reassignment took place, go to STEP 4, otherwise go to FIN.



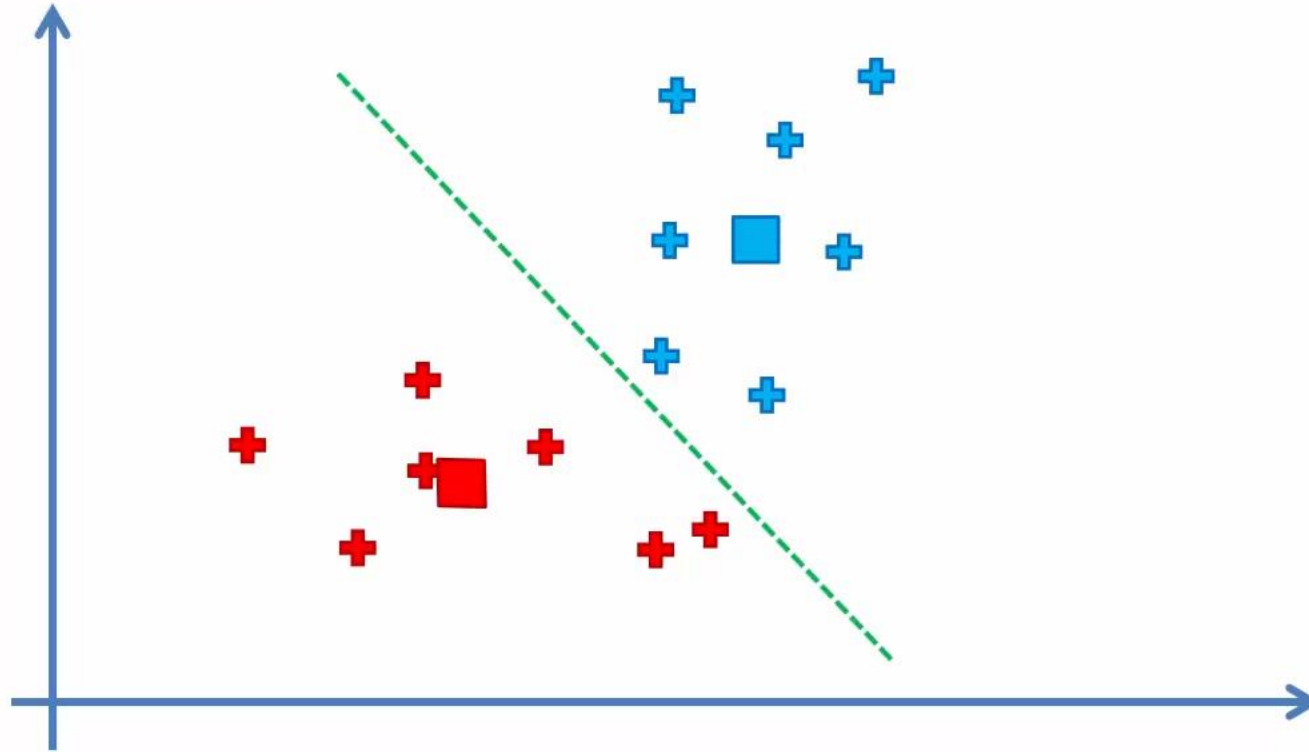
- The data point gets reassigned

STEP 4: Compute and place the new centroid of each cluster



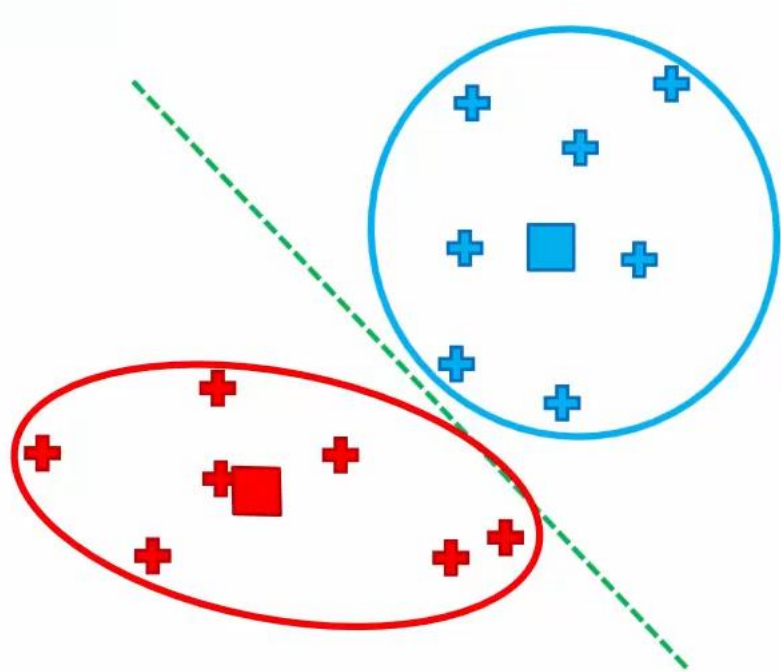
- Computing the new centroids for the clusters

STEP 5: Reassign each data point to the new closest centroid.
If any reassignment took place, go to STEP 4, otherwise go to FIN.

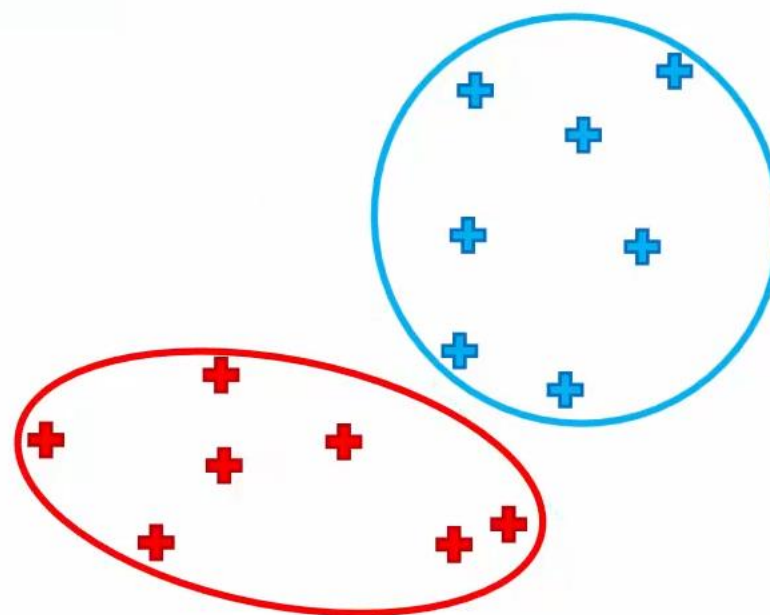


- Now, this time we do not need to reassign any data points.
- The algorithm has converged.

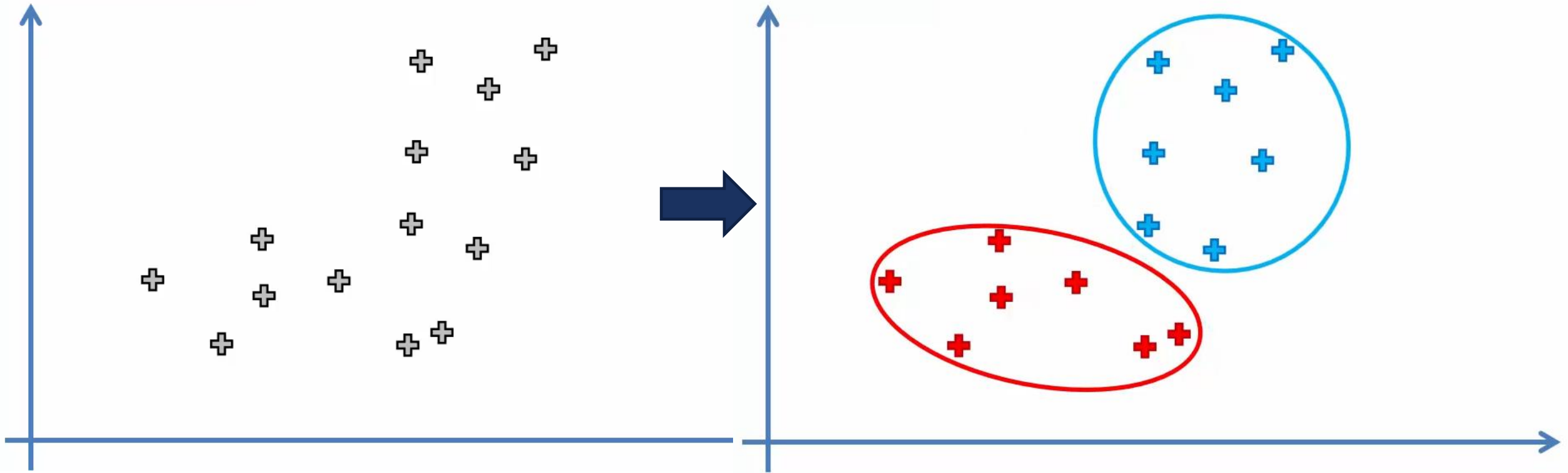
FIN: Your Model Is Ready



FIN: Your Model Is Ready



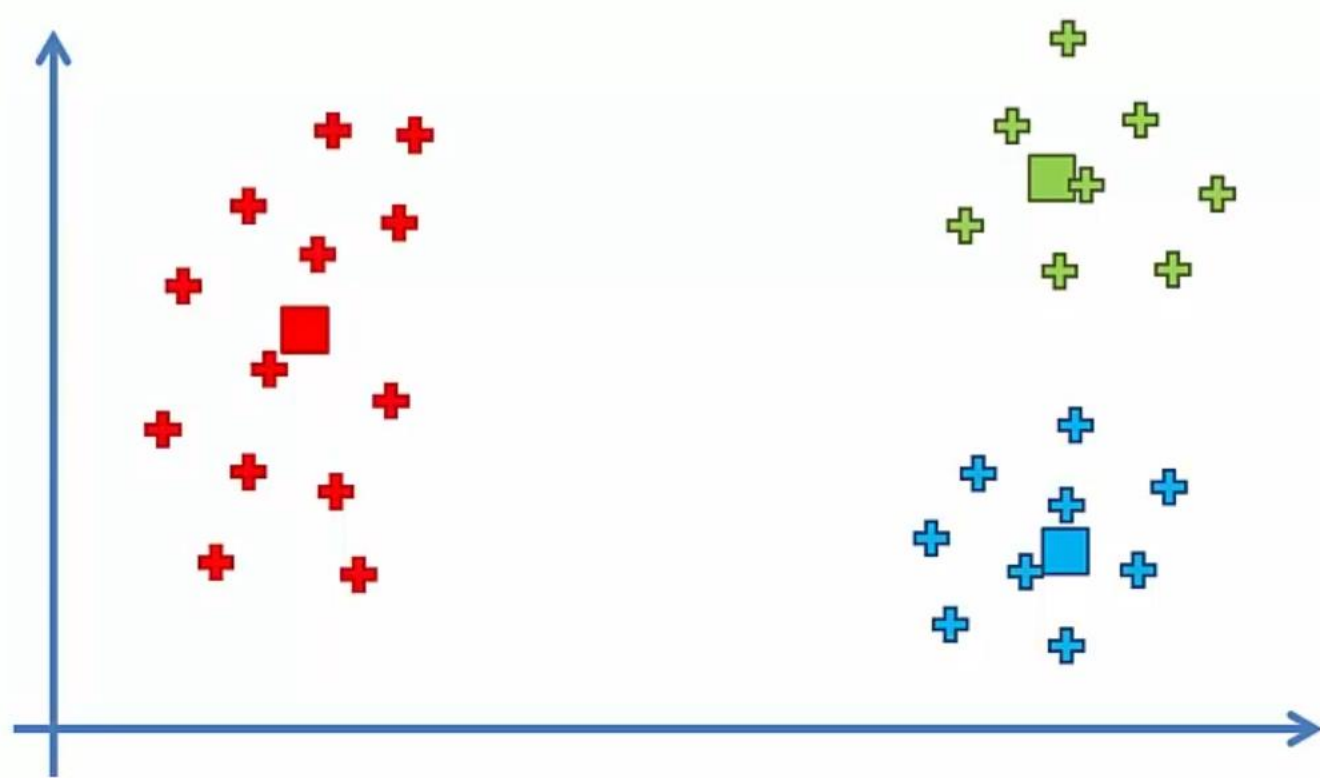
FIN: Your Model Is Ready



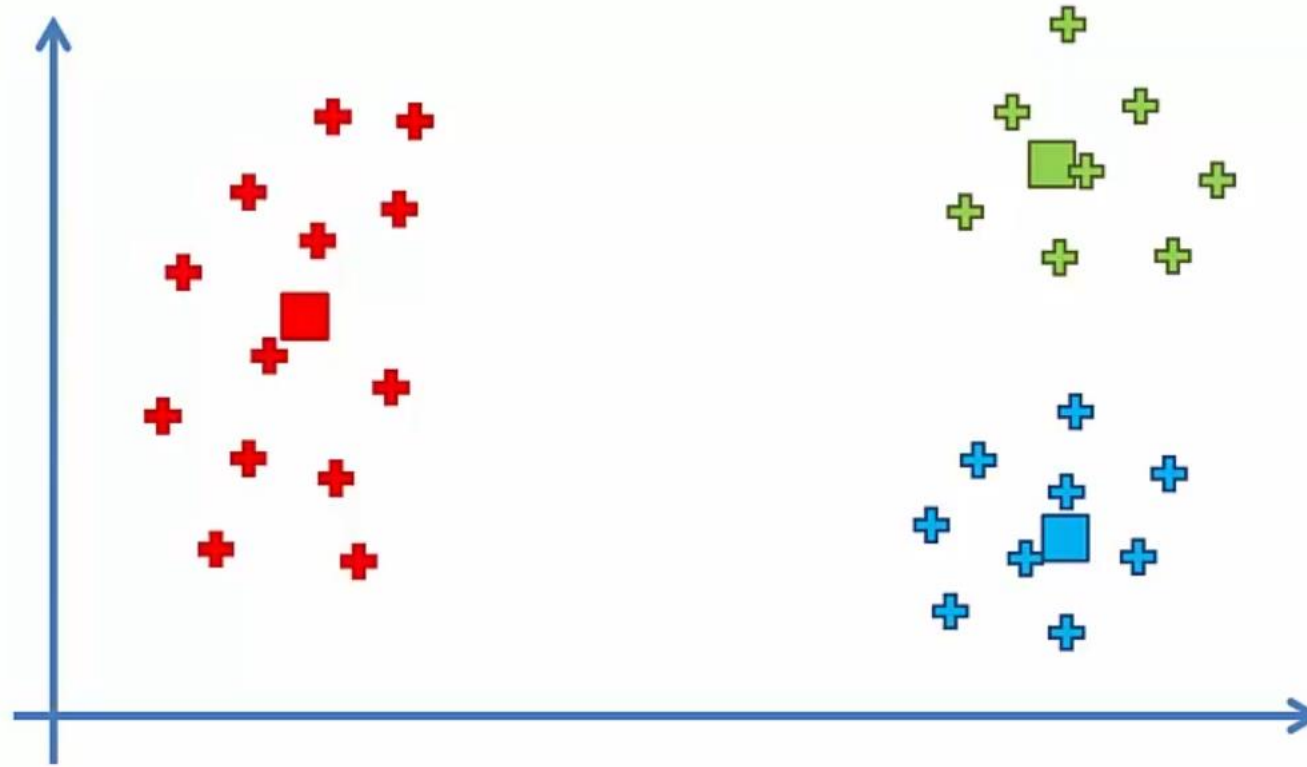
- From the initial data points to the clustered output.



...this correct random initialisation would lead us
to...



...the following three clusters



...the following three clusters

- Even if we move around the centroids a little bit, nothing is going to change

K-MEANS CLUSTERING: STEP-BY-STEP MANUAL SOLUTION

SOLVED EXAMPLE WITH DETAILED CALCULATIONS



DATASET

- We choose $K = 2$ clusters
- Points:
- $A(1,1)$
- $B(1.5,2)$
- $C(3,4)$
- $D(5,7)$
- $E(3.5,5)$
- $F(4.5,5)$

EUCLIDEAN DISTANCE FORMULA

- Distance Formula:
- $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Centroid Formula:
- Centroid = (Average x-values, Average y-values)

STEP 1: INITIALIZE CENTROIDS

- Choose initial centroids randomly:
- $C1 = A(1,1)$
- $C2 = D(5,7)$

STEP 2: DISTANCE CALCULATIONS (A AND B)

- Point A(1,1):
 - Distance to C1 = 0
 - Distance to C2 = $\sqrt{[(1-5)^2 + (1-7)^2]} = \sqrt{52} = 7.21$
 - Assign A to Cluster 1
- Point B(1.5,2):
 - Distance to C1 = $\sqrt{[(1.5-1)^2 + (2-1)^2]} = \sqrt{1.25} = 1.12$
 - Distance to C2 = $\sqrt{[(1.5-5)^2 + (2-7)^2]} = \sqrt{37.25} = 6.10$
 - Assign B to Cluster 1

DISTANCE CALCULATION FOR POINT C

- Point C(3,4):
- Distance to C1 = $\sqrt{(3-1)^2 + (4-1)^2}$
- = $\sqrt{4 + 9}$
- = $\sqrt{13} = 3.61$

- Distance to C2 = $\sqrt{(3-5)^2 + (4-7)^2}$
- = $\sqrt{4 + 9}$
- = $\sqrt{13} = 3.61$

- Equal distance from both centroids
- Assign C temporarily to Cluster 1

DISTANCE CALCULATIONS (D, E AND F)

- Point D(5,7):
 - Distance to C1 = 7.21
 - Distance to C2 = 0
 - Assign D to Cluster 2
- Point E(3.5,5):
 - Distance to C1 = $\sqrt{[(3.5-1)^2 + (5-1)^2]} = 4.72$
 - Distance to C2 = $\sqrt{[(3.5-5)^2 + (5-7)^2]} = 2.50$
 - Assign E to Cluster 2
- Point F(4.5,5):
 - Distance to C1 = 5.32
 - Distance to C2 = 2.06
 - Assign F to Cluster 2

CLUSTER ASSIGNMENT AFTER ITERATION 1

- Cluster 1: A(1,1), B(1.5,2), C(3,4)
- Cluster 2: D(5,7), E(3.5,5), F(4.5,5)

STEP 3: RECALCULATE CENTROIDS

- New centroid for Cluster 1:
 - $x = (1 + 1.5 + 3) / 3 = 1.83$
 - $y = (1 + 2 + 4) / 3 = 2.33$
 - New C1 = (1.83, 2.33)

- New centroid for Cluster 2:
 - $x = (5 + 3.5 + 4.5) / 3 = 4.33$
 - $y = (7 + 5 + 5) / 3 = 5.67$
 - New C2 = (4.33, 5.67)

STEP 4: CHECK AGAIN

- Recalculate distances using new centroids.
- All points remain in the same clusters.
- No changes occur in assignments.
- Algorithm converges.

FINAL RESULT

- Final Cluster 1 $\rightarrow$ A, B, C
- Final Cluster 2 $\rightarrow$ D, E, F
- K-Means stops when cluster assignments no longer change.

QUESTIONS

